

Topic 2: The Discrete Fourier Transform

DFT equation, basis functions, spectra, real/complex sinusoids, and inverse DFT

Digital Signal Processing for Sound and Music · Topic 2

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Lecture roadmap

1 DFT equation

from samples to frequency bins

2 DFT basis

complex exponentials as analysis signals

3 Interpretation

magnitude, phase, bin frequency

4 Sinusoids

complex and real sinusoid spectra

5 IDFT

reconstructing time-domain signals

6 Exercise 2

implement sinusoids, DFT, IDFT, and magnitude spectrum

The DFT: a change of viewpoint

Input: a finite sequence of samples

The signal is known only at N discrete time positions.

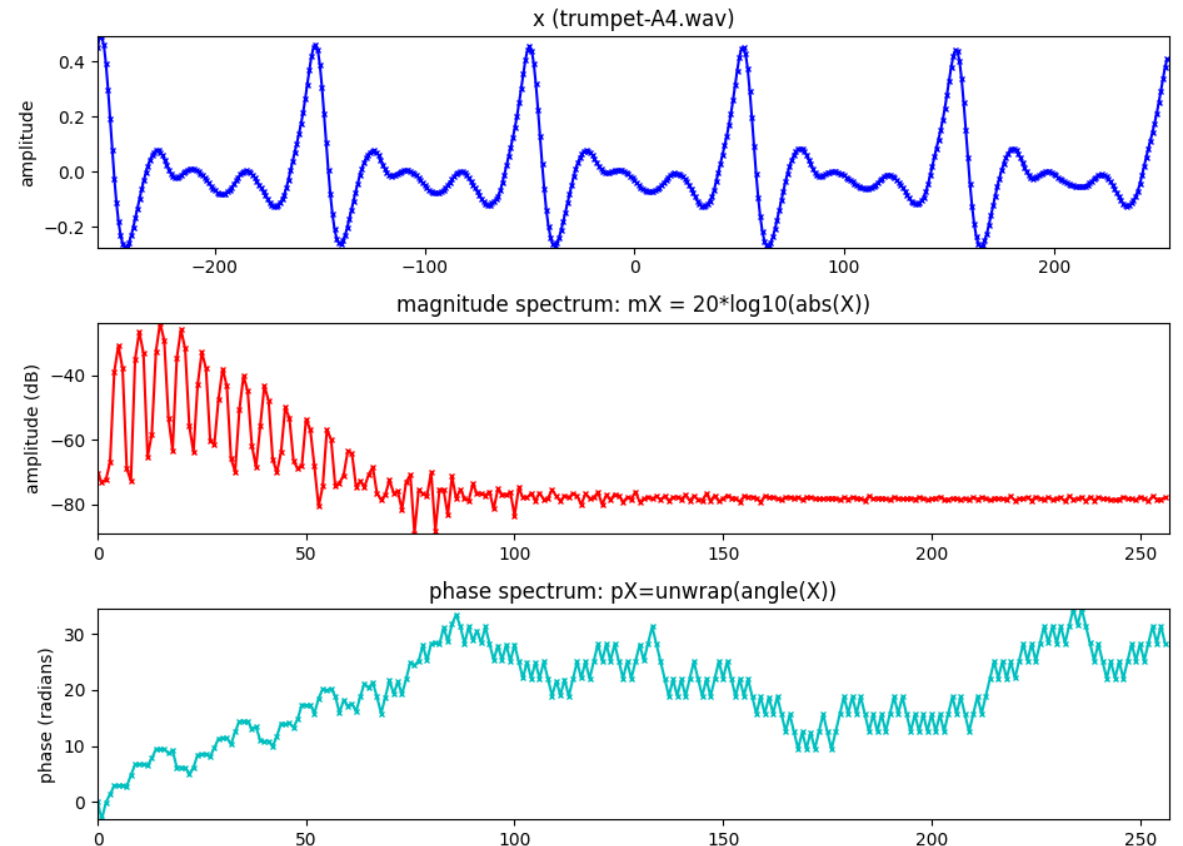
The DFT compares it against N analysis sinusoids.

The result is a complex number for each frequency bin.

Output: a spectrum

Magnitude tells how much of a bin frequency is present.

Phase tells how that sinusoidal component is aligned in time.



DFT equation and index meanings

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \quad k = 0, \dots, N-1$$

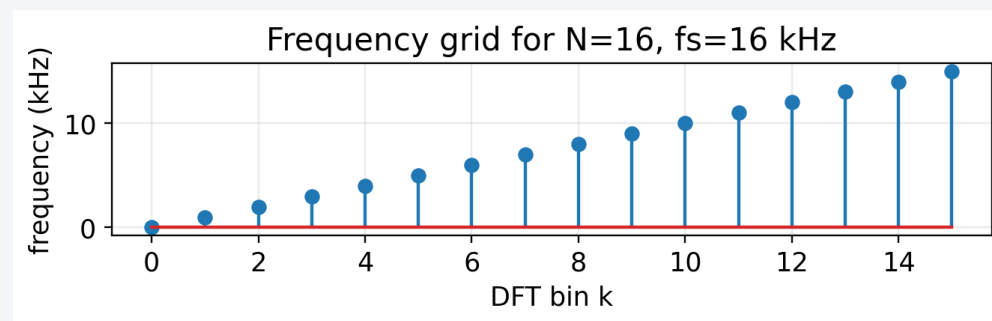
n : discrete time index(normalized time, $T = 1$)

k : discrete frequency index

$\omega_k = 2\pi k/N$: frequency in radians

$f_k = f_s k/N$: frequency in Hz(f_s : sampling rate)

Frequency index k maps each DFT bin to radians and to Hz.



Frequency bins are a fixed grid

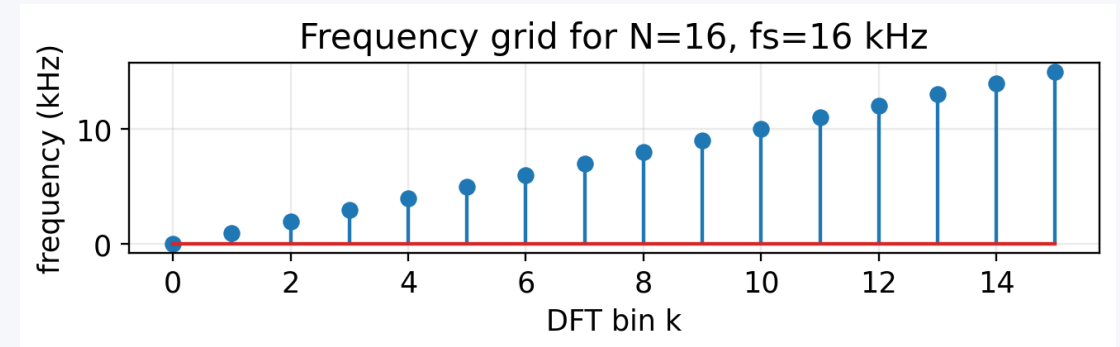
Bin frequency

$$f_k = \frac{k}{N}f_s, \quad \Delta f = \frac{f_s}{N}, \quad \omega_k = \frac{2\pi k}{N}$$

The DFT does not test all possible frequencies. It tests exactly N frequencies determined by N and f_s .

Increasing N gives a finer frequency grid.

A sinusoid exactly on a bin gives a sharp spectral peak; off-bin frequencies spread across bins.



DFT basis functions: complex exponentials

$$s_k^* = e^{-j2\pi kn/N} = \cos(2\pi kn/N) - j\sin(2\pi kn/N)$$

for $N = 4$, thus for $n = 0,1,2,3$; $k = 0,1,2,3$

$$s_0^* = \cos(2\pi \times 0 \times n/4) - j\sin(2\pi \times 0 \times n/4) = [1, 1, 1, 1]$$

$$s_1^* = \cos(2\pi \times 1 \times n/4) - j\sin(2\pi \times 1 \times n/4) = [1, -j, -1, j]$$

$$s_2^* = \cos(2\pi \times 2 \times n/4) - j\sin(2\pi \times 2 \times n/4) = [1, -1, 1, -1]$$

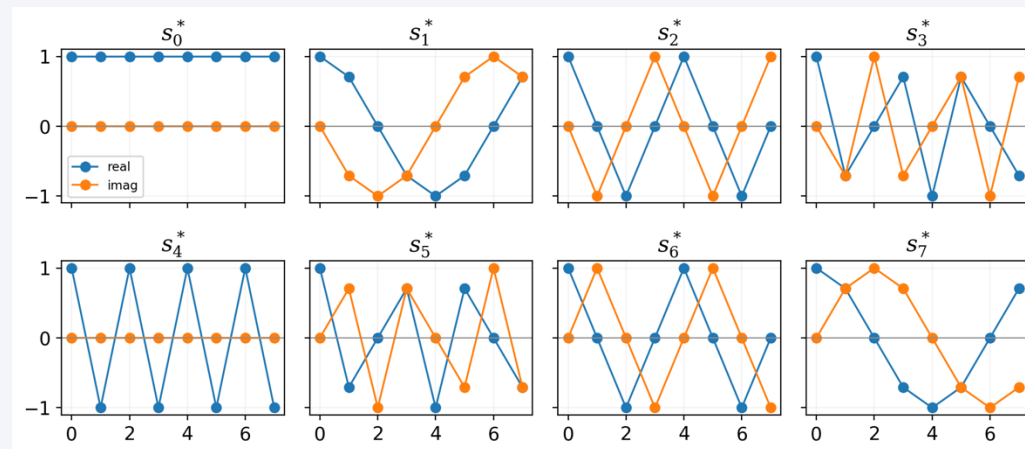
$$s_3^* = \cos(2\pi \times 3 \times n/4) - j\sin(2\pi \times 3 \times n/4) = [1, j, -1, -j]$$

Why these signals?

For each k , the DFT uses one complex exponential.

The conjugate basis s^*_k rotates in the opposite direction.

For $N = 4$ the four basis sequences can be written explicitly.

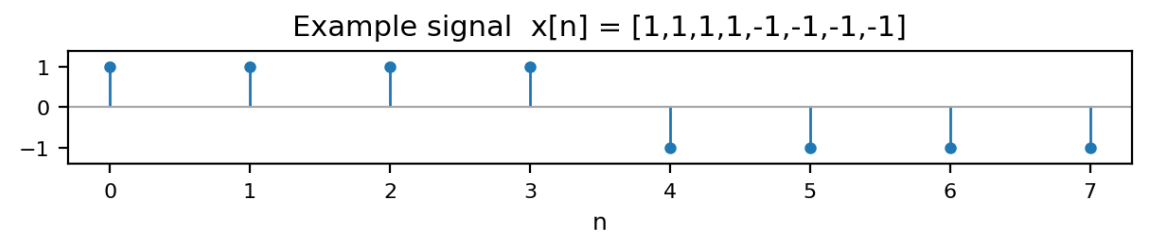


Plots for $N = 8$.

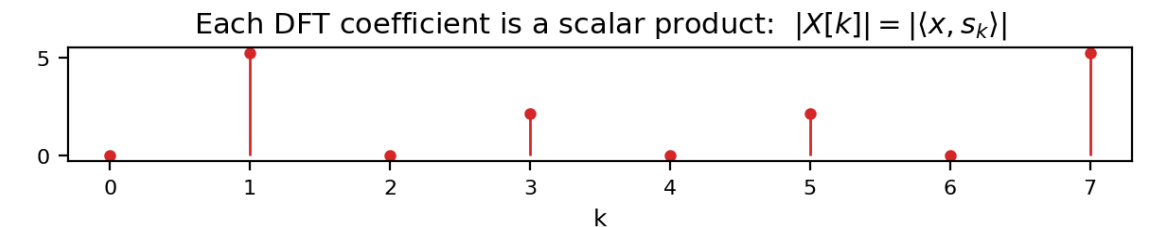
The DFT as scalar products

$$\langle x, s_k \rangle = \sum_{n=0}^{N-1} x[n] s_k^*[n] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

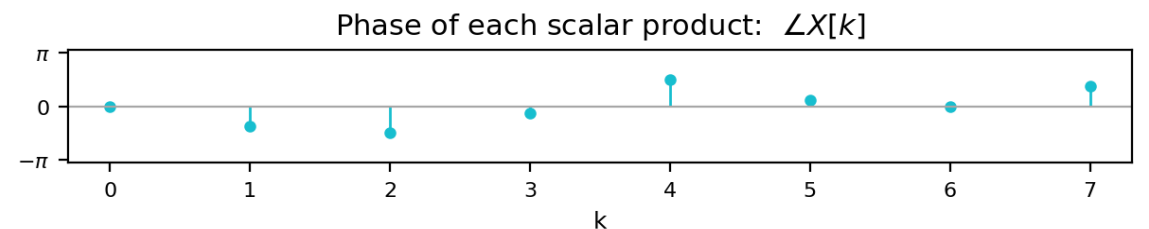
example signal $x[n]$



magnitude $\text{abs}(X[k])$



phase angle $\angle(X[k])$



Example: N = 4 scalar products

Worked example directly matched to the equations

$$x[n] = [1, -1, 1, -1]; N = 4$$

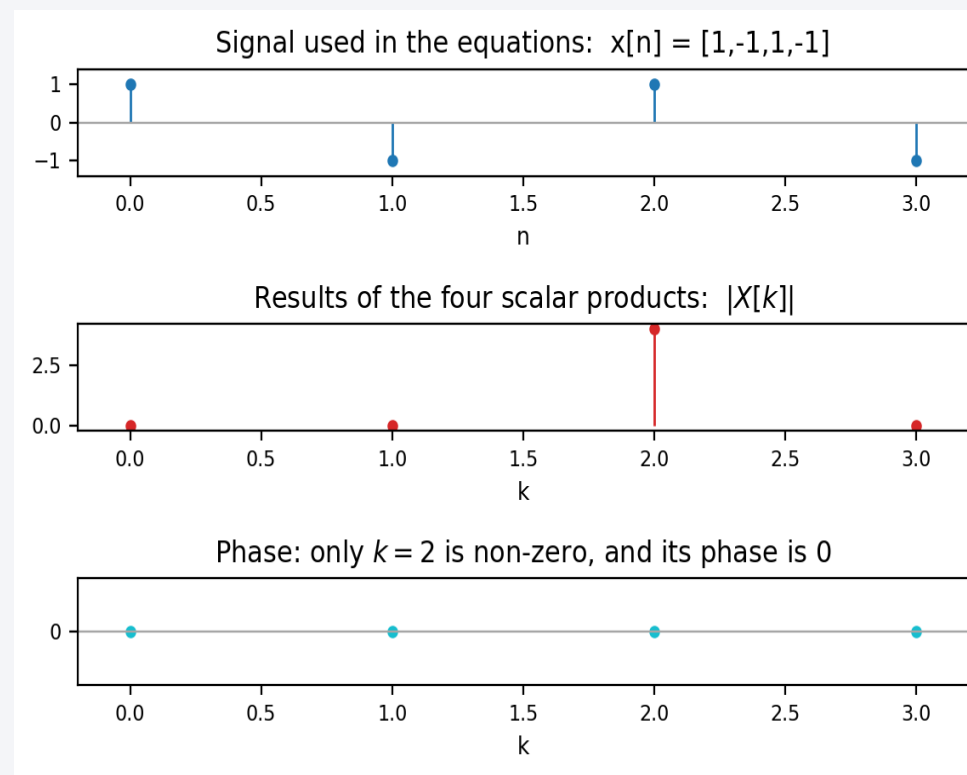
$$\langle x, s_0 \rangle = 1 \times 1 + (-1) \times 1 + 1 \times 1 + (-1) \times 1 = 0$$

$$\langle x, s_1 \rangle = 1 \times 1 + (-1) \times (-j) + 1 \times (-1) + (-1) \times j = 0$$

$$\langle x, s_2 \rangle = 1 \times 1 + (-1) \times (-1) + 1 \times 1 + (-1) \times (-1) = 4$$

$$\langle x, s_3 \rangle = 1 \times 1 + (-1) \times j + 1 \times (-1) + (-1) \times (-j) = 0$$

The plot shows the same result as the equations: $X[0]=0$, $X[1]=0$, $X[2]=4$, $X[3]=0$.



Magnitude and phase: what does the spectrum store?

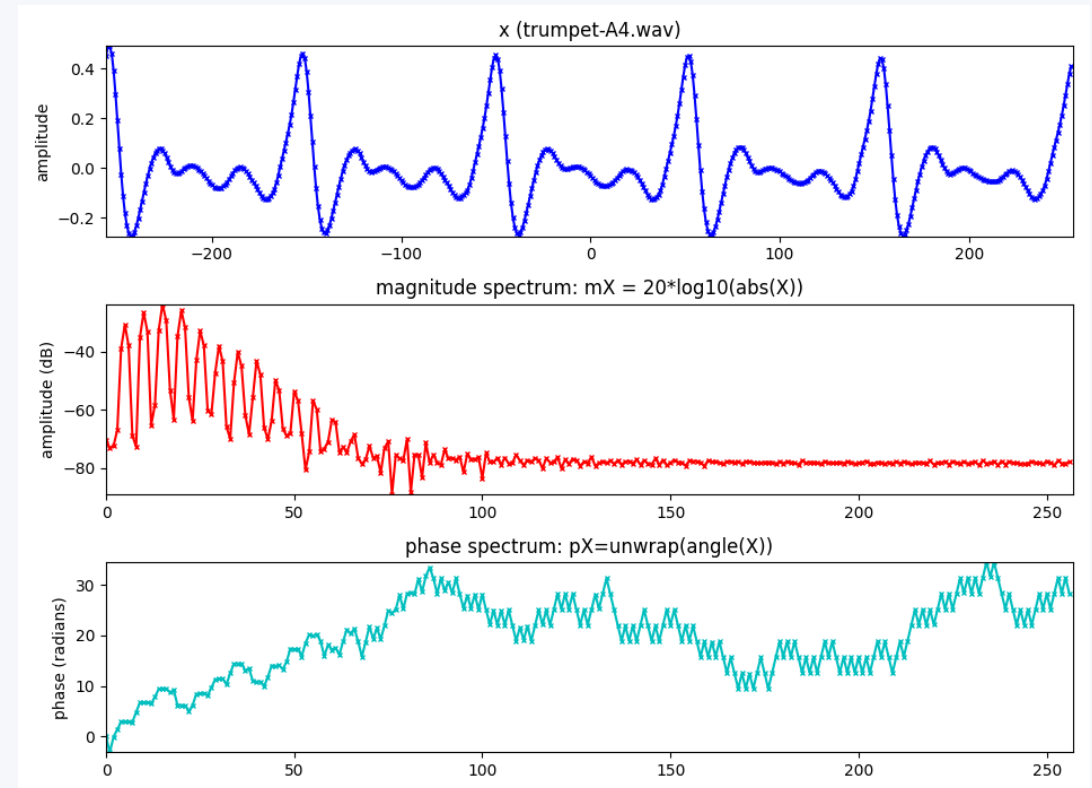
A DFT value is complex:

$$X[k] = |X[k]|e^{j\angle X[k]}$$

Magnitude $|X[k]|$: strength of the k th frequency component.

Phase $\angle X[k]$: temporal alignment of that component.

For audio, magnitude is often easier to interpret; phase is essential for reconstruction.



DFT of a complex sinusoid exactly on a bin

$$x_1[n] = e^{j2\pi k_0 n/N} \text{ for } n = 0, \dots, N-1$$

$$X_1[k] = \sum_{n=0}^{N-1} x_1[n] e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} e^{j2\pi k_0 n/N} e^{-j2\pi kn/N}$$

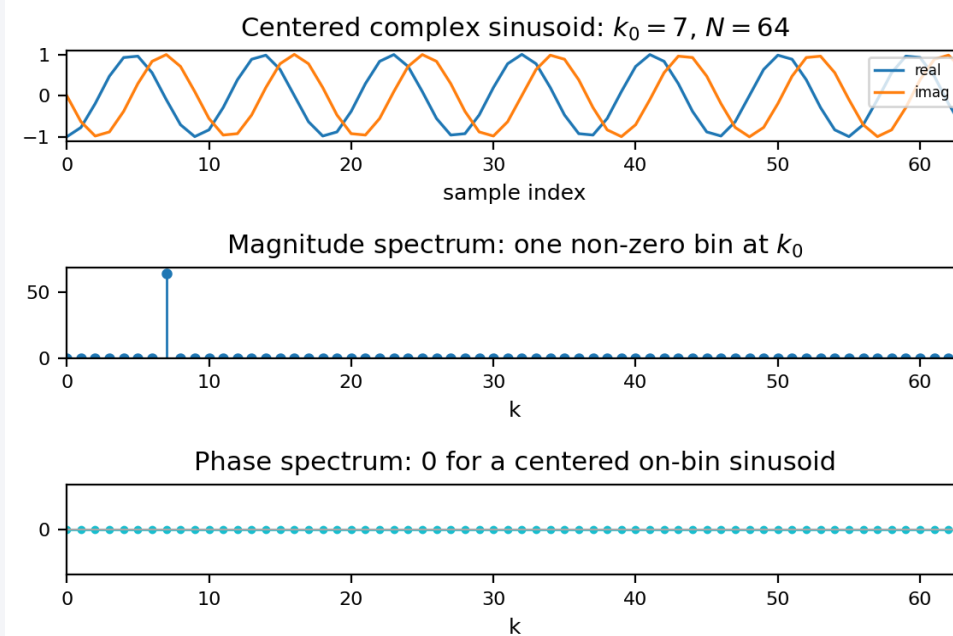
$$= \sum_{n=0}^{N-1} e^{-j2\pi(k-k_0)n/N}$$

$$= \frac{1 - e^{-j2\pi(k-k_0)}}{1 - e^{-j2\pi(k-k_0)/N}} \text{ (sum of a geometric series)}$$

if $k \neq k_0$, denominator $\neq 0$ and numerator $= 0$
thus $X_1[k] = 0$ for $k \neq k_0$ and $X_1[k_0] = N$

Main result

All energy is at bin k_0 .
For the centered signal, the phase is 0.



DFT of any complex sinusoid

$$x_2[n] = e^{j(2\pi f_0 n + \phi)} \text{ for } n = 0, \dots, N-1$$

$$X_2[k] = \sum_{n=0}^{N-1} x_2[n] e^{-j2\pi kn/N}$$

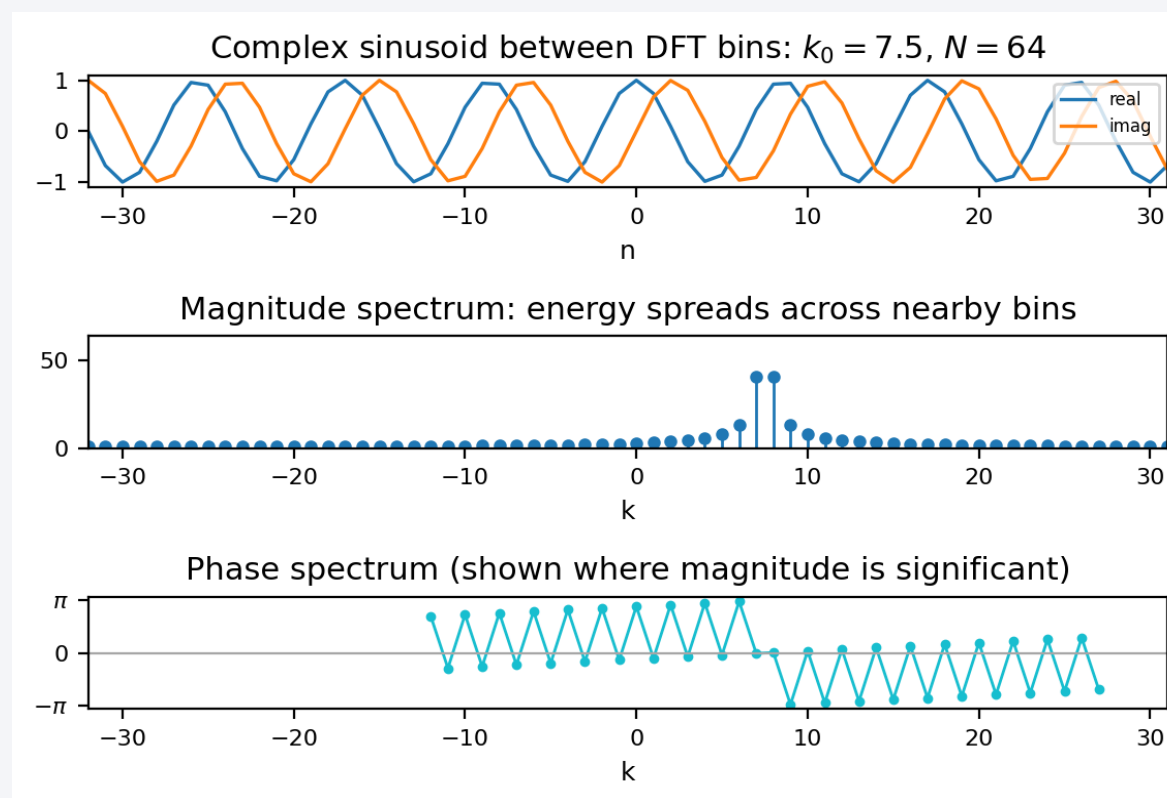
$$= \sum_{n=0}^{N-1} e^{j(2\pi f_0 n + \phi)} e^{-j2\pi kn/N}$$

$$= e^{j\phi} \sum_{n=0}^{N-1} e^{-j2\pi(k/N - f_0)n}$$

$$= e^{j\phi} \frac{1 - e^{-j2\pi(k/N - f_0)N}}{1 - e^{-j2\pi(k/N - f_0)}}$$

Off-bin case

Off-bin sinusoids spread energy across bins.



DFT of real sinusoids

$$x_3[n] = A_0 \cos(2\pi k_0 n/N) = \frac{A_0}{2} e^{j2\pi k_0 n/N} + \frac{A_0}{2} e^{-j2\pi k_0 n/N}$$

$$X_3[k] = \sum_{n=-N/2}^{N/2-1} x_3[n] e^{-j2\pi kn/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \left(\frac{A_0}{2} e^{j2\pi k_0 n/N} + \frac{A_0}{2} e^{-j2\pi k_0 n/N} \right) e^{-j2\pi kn/N}$$

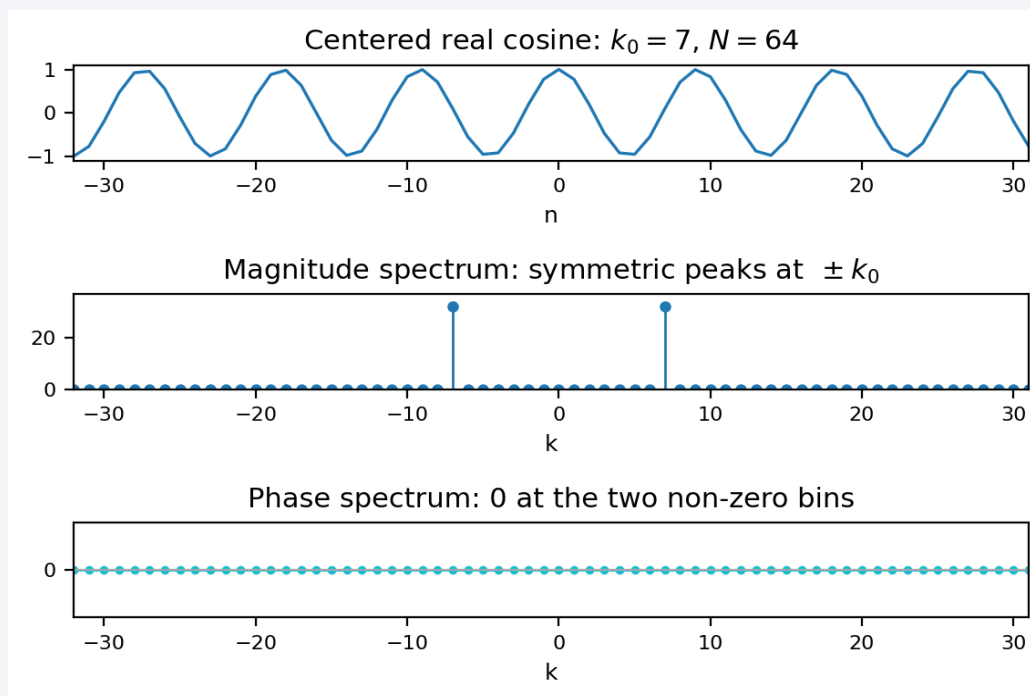
$$= \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{j2\pi k_0 n/N} e^{-j2\pi kn/N} + \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi k_0 n/N} e^{-j2\pi kn/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi(k-k_0)n/N} + \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi(k+k_0)n/N}$$

$$= N \frac{A_0}{2} \text{ for } k = k_0, -k_0; 0 \text{ for rest of } k$$

Positive + negative frequencies

A centered real cosine has two symmetric peaks. Their phases are 0.



Inverse DFT: from spectrum back to samples

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] s_k[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$
$$n = 0, 1, \dots, N-1$$

$$X[k] = [0, 4, 0, 0]; N = 4$$

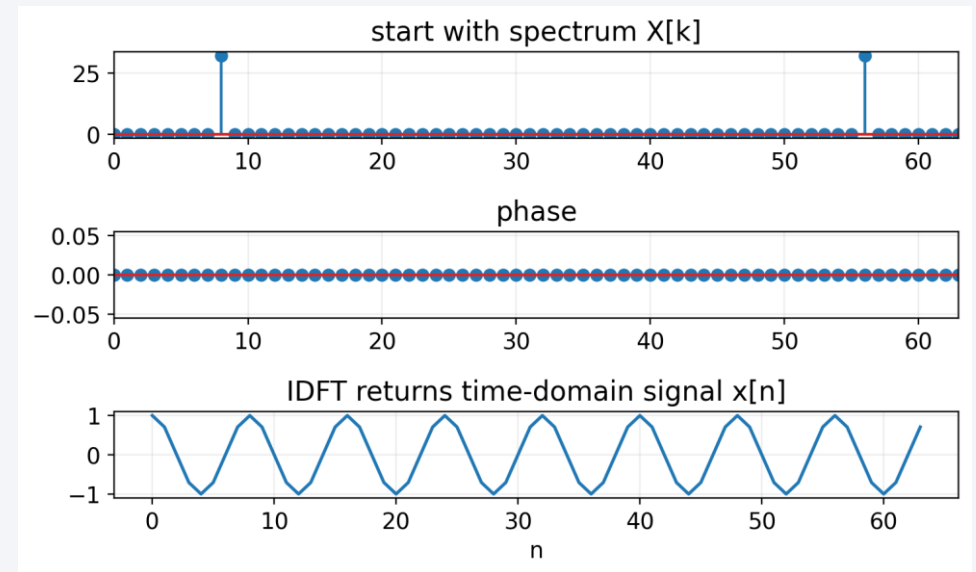
$$x[0] = \frac{1}{4} (X * s)[n=0] = \frac{1}{4} (0 * 1 + 4 * 1 + 0 * 1 + 0 * 1) = 1$$

$$x[1] = \frac{1}{4} (X * s)[n=1] = \frac{1}{4} (0 * 1 + 4 * j + 0 * (-1) + 0 * (-j)) = j$$

$$x[2] = \frac{1}{4} (X * s)[n=2] = \frac{1}{4} (0 * 1 + 4 * (-1) + 0 * 1 + 0 * (-1)) = -1$$

$$x[3] = \frac{1}{4} (X * s)[n=3] = \frac{1}{4} (0 * 1 + 4 * (-j) + 0 * (-1) + 0 * j) = -j$$

The IDFT reconstructs $x[n]$ by summing all basis functions weighted by $X[k]$.

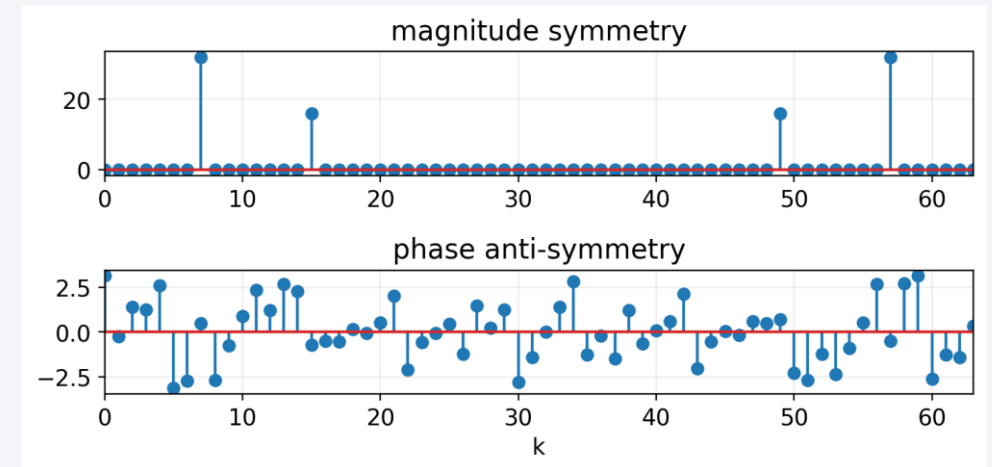


Inverse DFT for real signals

$$X[k] = |X[k]|e^{j\angle X[k]} \text{ and } X[-k] = |X[k]|e^{-j\angle X[k]} \\ \text{for } k = 0, 1, \dots, N/2$$

Conjugate symmetry

For real-valued time signals, positive and negative frequency coefficients are complex conjugates: equal magnitudes and opposite phases.

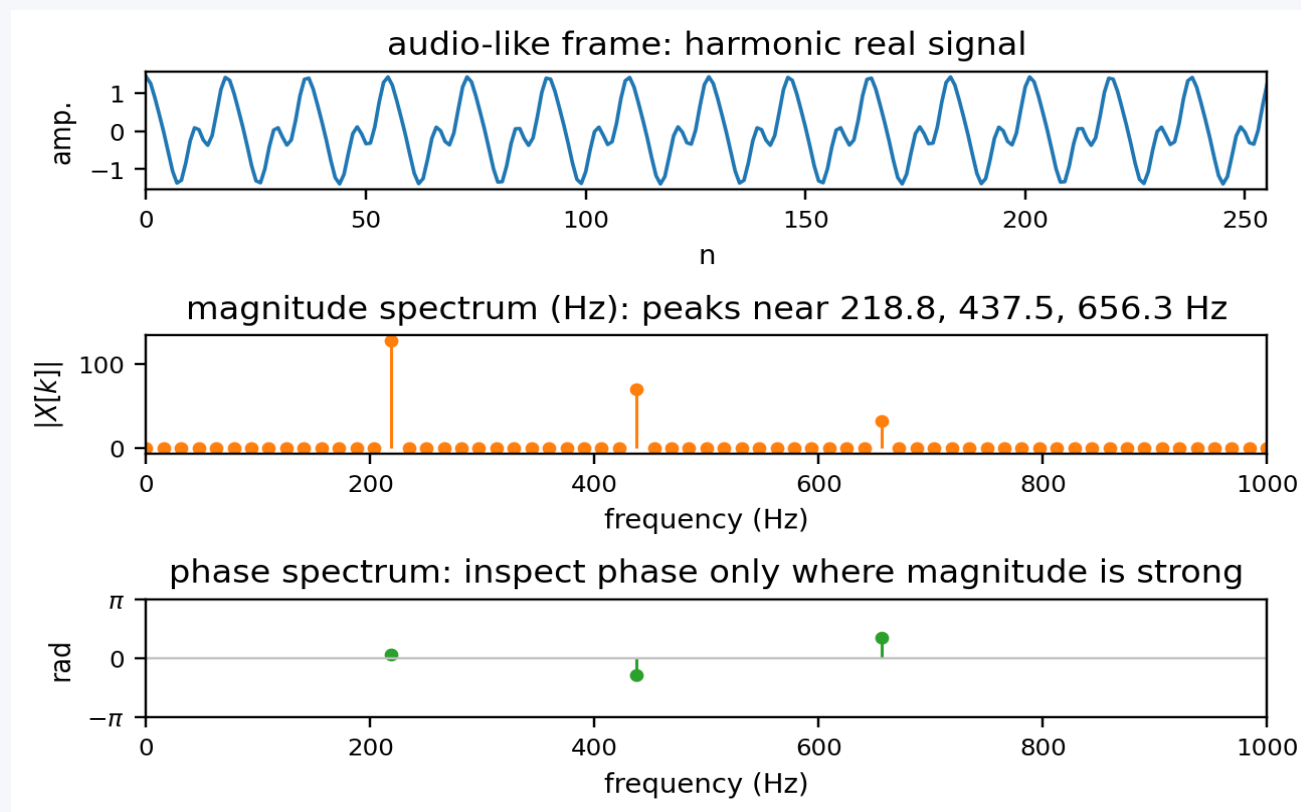


Practical interpretation for audio signals

When reading a spectrum, ask:

- What is the sampling rate f_s ?
- What is the DFT size N ?
- Which bin k corresponds to the frequency I care about?
- Is the sinusoid exactly at a bin center?
- Am I looking at magnitude, phase, or both?

These questions determine whether a peak is easy to interpret, smeared across bins, or mirrored by real-signal symmetry.



Example notebook: DFT analysis / resynthesis workflow

examples/dft-examples.ipynb turns the DFT equation into a concrete pipeline: read a sound, analyze one frame, and synthesize it back.

1 Load sound

read audio and listen

2 Select frame

choose a short segment

3 Window + DFT

compute magnitude +
phase

4 IDFT synth

reconstruct the frame

Key code block

```
window = 'hamming'
M = 256      # analysis-window size
N = 1024     # DFT size
frame_time = 0.2

w = get_window(window, M)
x_frame = x[start_sample:start_sample + M]
mX, pX = DFT.dftAnal(x_frame, w, N)
y_frame = DFT.dftSynth(mX, pX, w.size)
```

What students should notice

- M controls how much time context is analyzed.
- N controls the spacing between DFT frequency bins.
- mX and pX are the magnitude and phase spectra.
- Both magnitude and phase are needed for resynthesis.

What the DFT example shows

The notebook links abstract DFT coefficients to visible and audible results: a selected frame, its spectrum, and a resynthesized waveform.

1 • Analysis

Isolate a short frame, multiply it by a window, and transform it into magnitude and phase.

2 • Interpretation

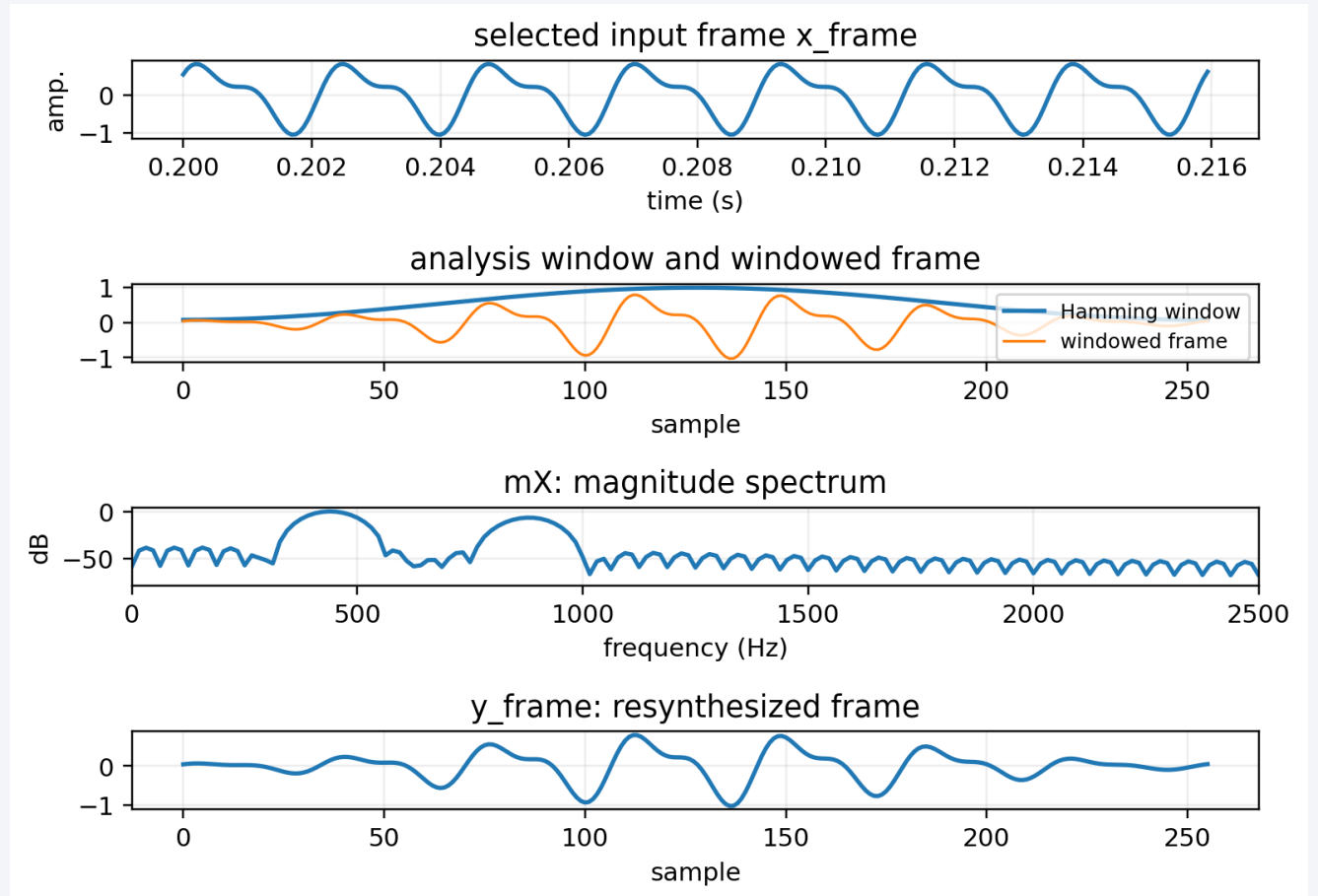
Peaks in mX indicate strong sinusoidal components; pX gives their phase alignment.

3 • Resynthesis

Use the DFT coefficients to reconstruct the time-domain frame.

4 • Parameters

M fixes the frame duration; N fixes the DFT frequency grid used to analyze it.



Exercise 2: Sinusoids and the DFT

Second lab: turn DFT formulas into code, plots, spectra, and reconstruction.

What you will practice

- generate and inspect real sinusoids
- build DFT complex sinusoids for bin k
- implement DFT, IDFT, and magnitude spectra
- connect spectrum peaks with the signal you created

How to start

Open `exercises/E2-Sinusoids-and-DFT.ipynb`

Use: NumPy · Matplotlib · complex exponentials · plots

Compare with: `examples/dft-examples.ipynb`

Lab structure

1 · Real sinusoid

- A , f , ϕ , f_s , duration
- time plot + listening

2 · DFT basis

- complex sinusoid
- bin k , length N

3 · DFT

- scalar products
- one $X[k]$ per bin

4 · IDFT

- sum weighted bases
- include $1/N$ scaling

5 · Spectrum

- magnitude peaks
- interpret bin locations

What to remember from Topic 2

1

DFT equation

computes one complex coefficient per frequency bin

2

Basis functions

complex exponentials indexed by k

3

Scalar products

measure similarity between $x[n]$ and each basis function

4

Complex sinusoids

one bin if on-grid, leakage if off-grid

5

Real sinusoids

two symmetric peaks

6

IDFT

reconstructs samples by summing all components

References and credits

- MTG sms-tools-materials repository: lecture slides, examples, exercises, and sounds: <https://github.com/MTG/sms-tools-materials>
- Exercise 2 notebook: E2-Sinusoids-and-DFT.ipynb
- Julius O. Smith, Mathematics of the Discrete Fourier Transform: <https://ccrma.stanford.edu/~jos/mdft/>
- Discrete Fourier Transform overview: https://en.wikipedia.org/wiki/Discrete_Fourier_transform
- Sounds from Freesound and sms-tools-materials sounds folder.